

The pH paper contains a mixture of indicators which gives different colours across the entire pH range. It is used to measure pH of the sample.

Key Concept

The quantity of strong acid or base that is added to change the pH of one liter of buffer solution by one pH unit is known as **buffer capacity**.

1.5.3 Henderson–Hasselbalch equation

An equation showing the relationship between a buffer's pH and the relative amounts of the buffer's weak acid and its conjugate base is called as Henderson-Hasselbalch equation. Consider the dissociation of a weak acid (HA). At equilibrium,

$$HA \rightleftharpoons H^+ + A^-$$

The dissociation constant (K_a) is,

$$K_a = \frac{[H^+]\,[A^-]}{[HA]} \qquad \text{.....(1.1)}$$

Rearrange the equation 1.1 for hydrogen ion concentration

$$[H^+] = K_a \frac{[HA]}{[A^-]} \qquad \text{.....(1.2)}$$

Taking the logarithm to base 10, on both side of the equation 1.2,

$$\log_{10}[H^+] = \log_{10} K_a + \log_{10} \frac{[HA]}{[A^-]} \qquad \text{.....(1.3)}$$

By substituting pH and pK_a values to the equation 1.3

$$-\text{pH} = -\text{p}K_a + \log_{10} \frac{[HA]}{[A^-]} \qquad \text{.....(1.4)}$$

Multiplying the equation 1.4 by (–1)

$$\text{pH} = \text{p}K_a - \log_{10} \frac{[HA]}{[A^-]} \qquad \text{.....(1.5)}$$

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